

‘Hey mum, I dropped my phone down the toilet’: Investigating Hi Mum and Dad SMS Scams in the United Kingdom

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Microsoft Research Pre-USENIX Security’25

12th Aug 2025

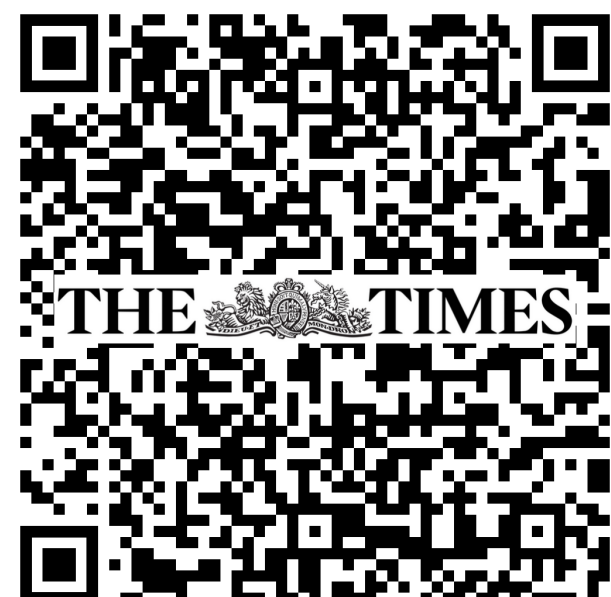
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‘Hi Mum and Dad’ ... inside the emotive text scam

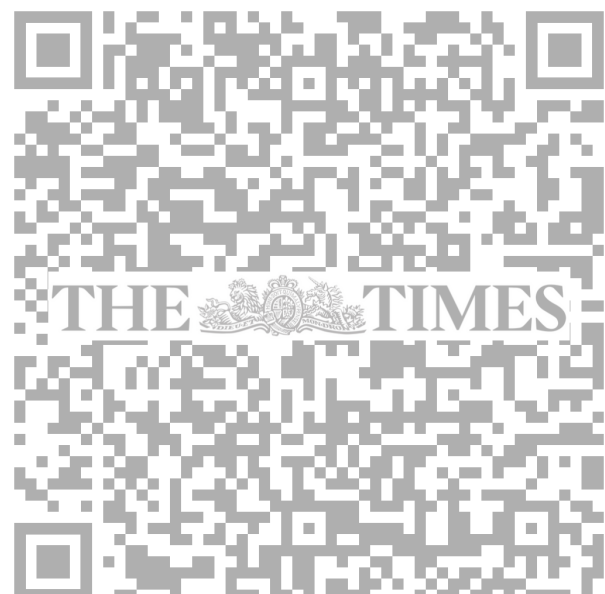
Fraudsters are posing as vulnerable children to trick parents out of thousands of pounds — but the police and phone companies are playing catch-up

Full Article:

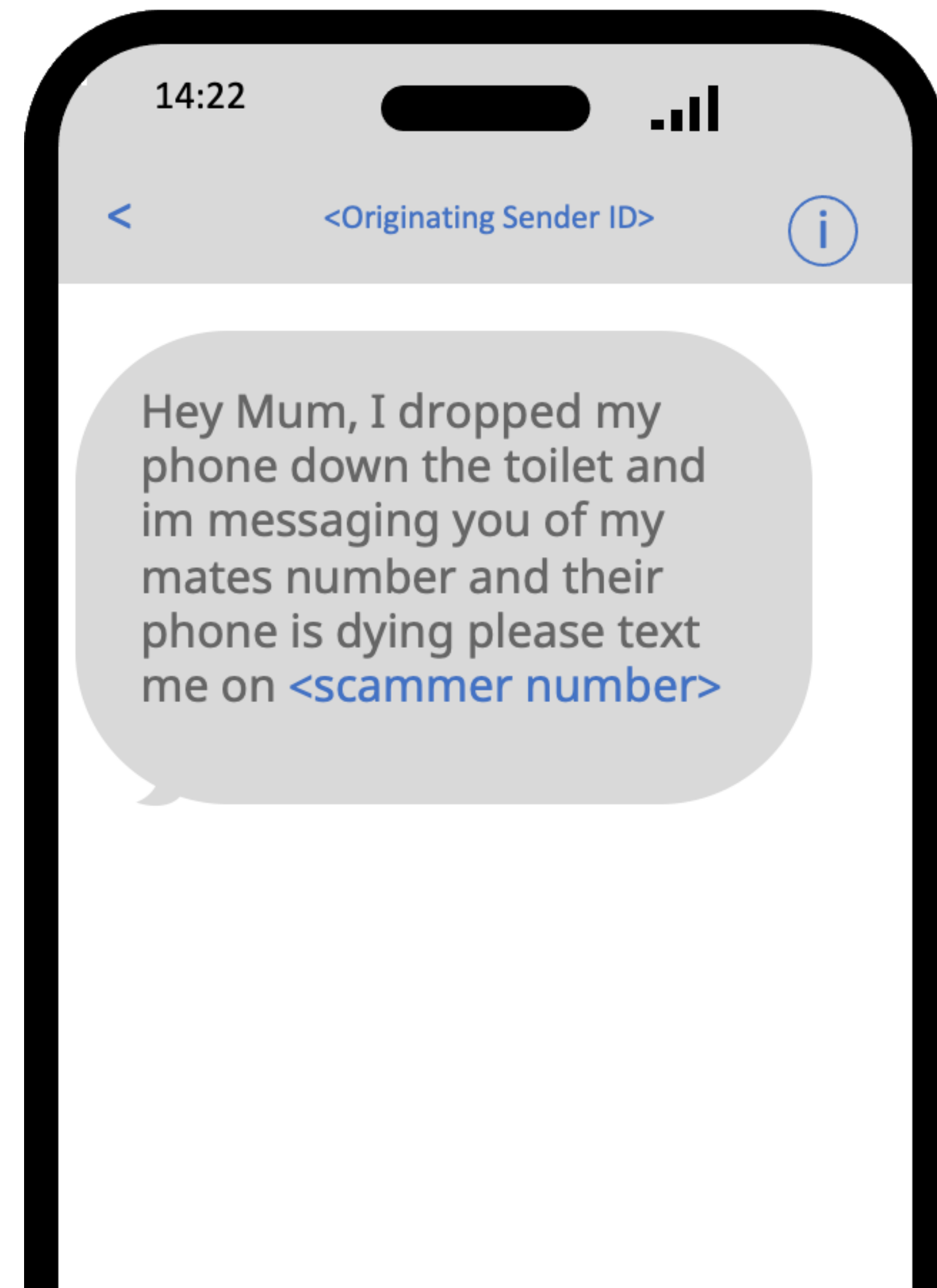


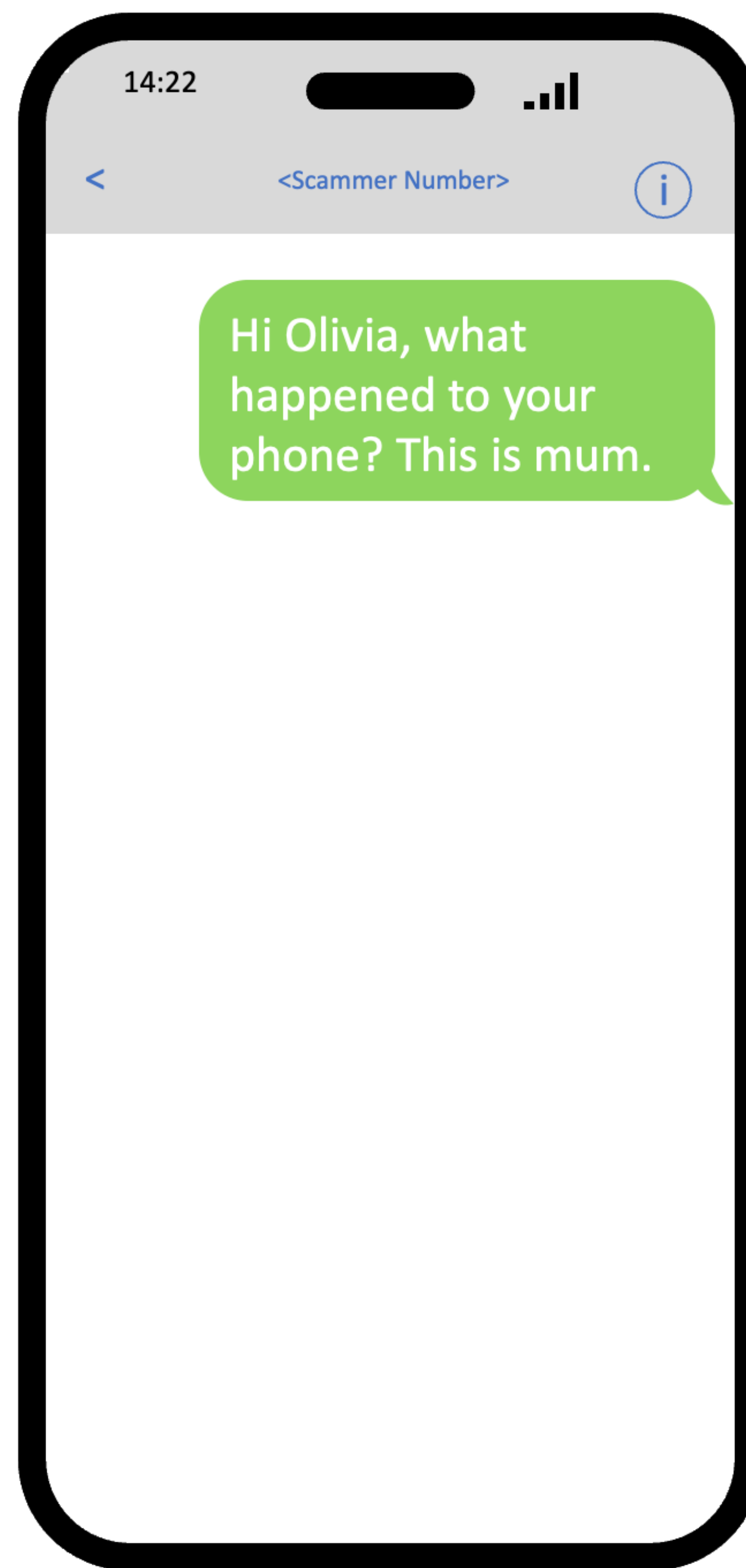
‘Hi Mum and Dad’ ... inside the emotive text scam

Fraudsters are posing as vulnerable children to trick parents out of thousands of pounds — but the police and phone companies are playing catch-up



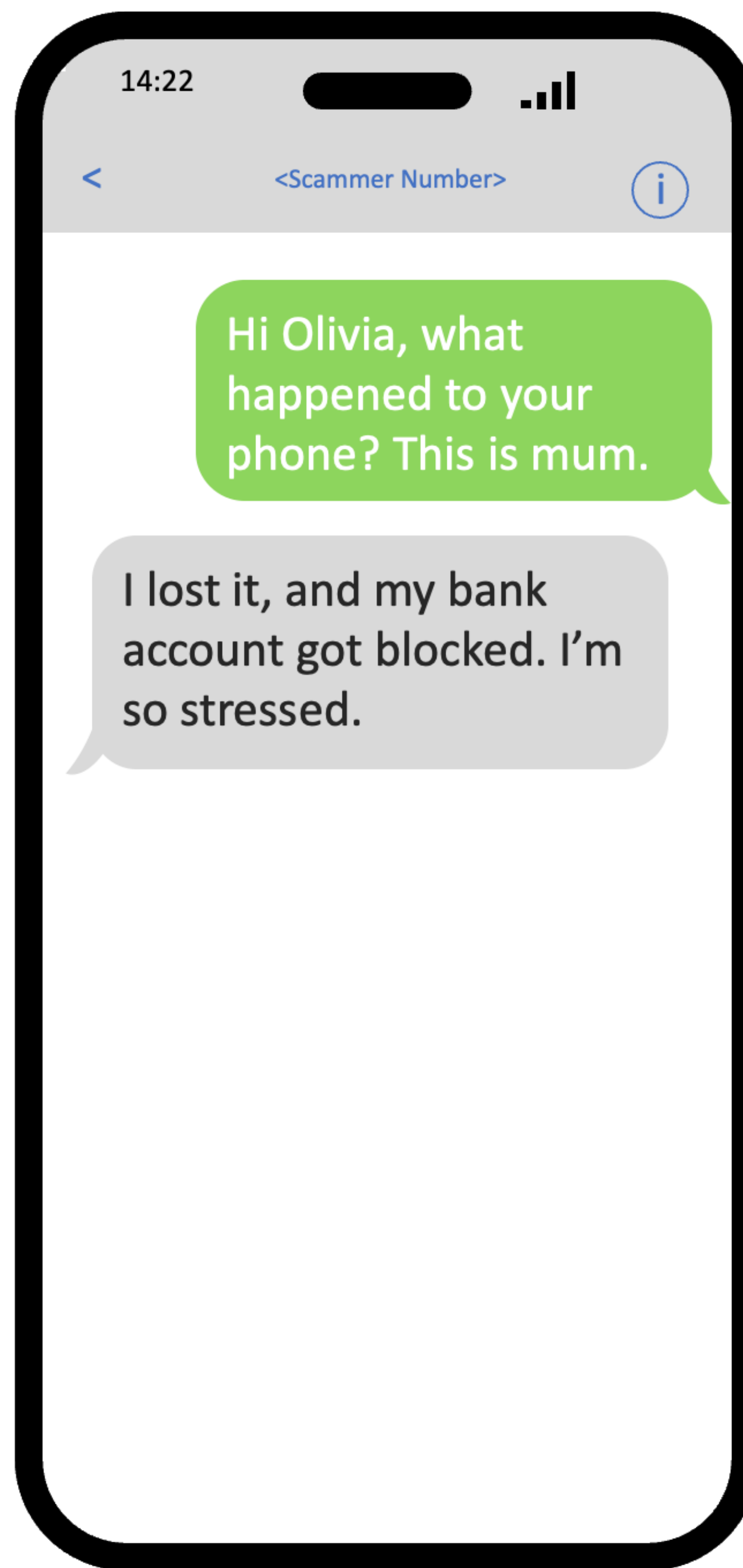
Initial Scam Text

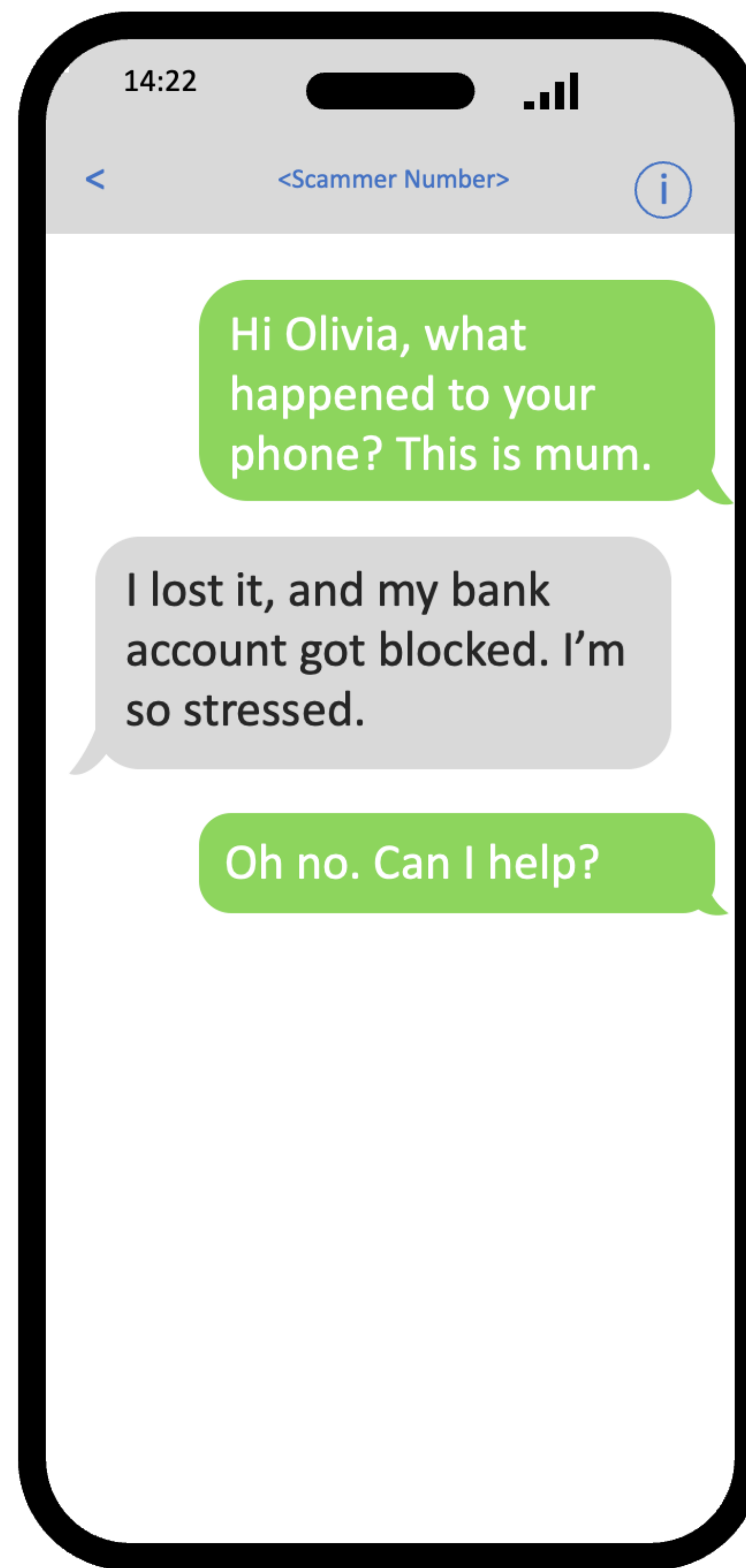




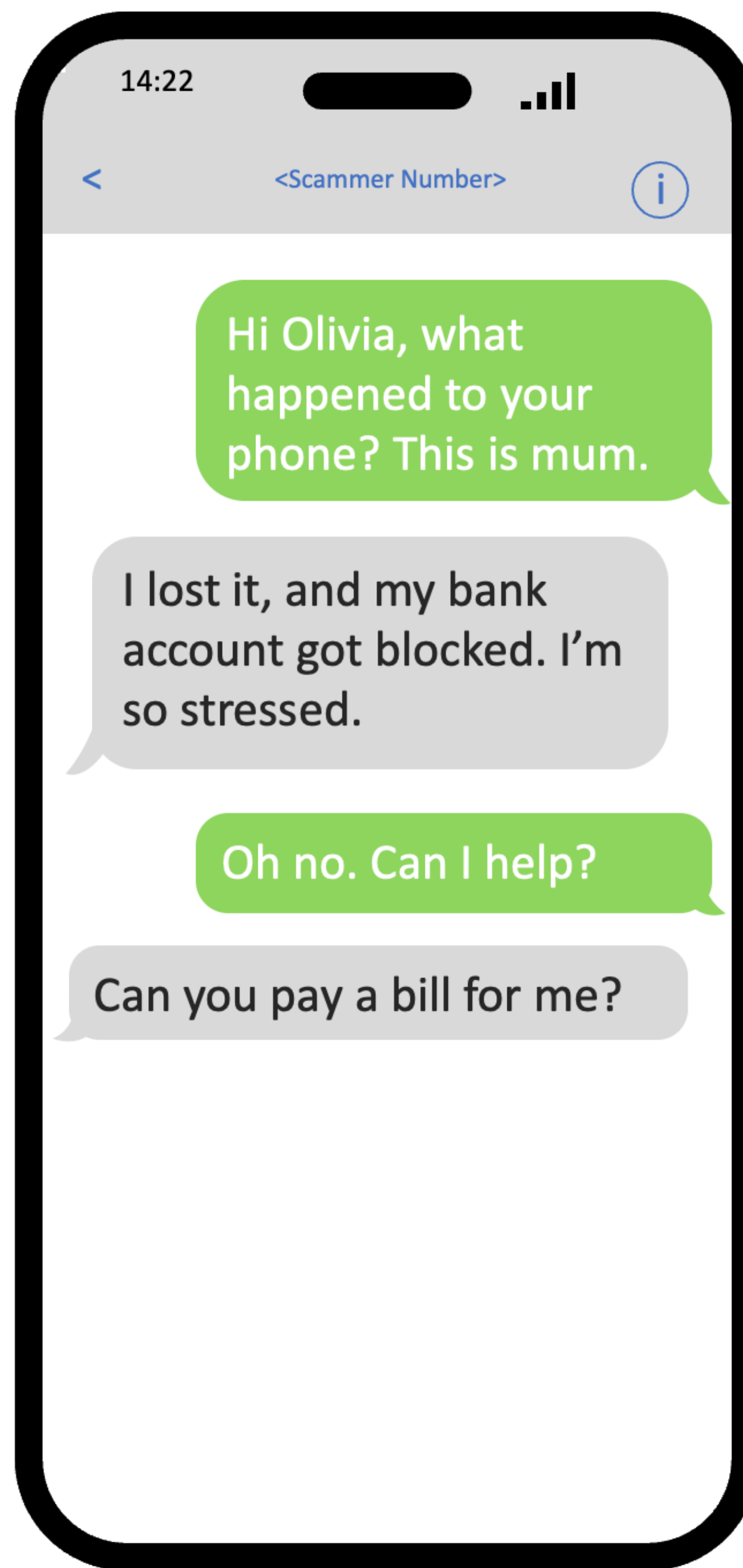
We initiate conversation with
scammer number

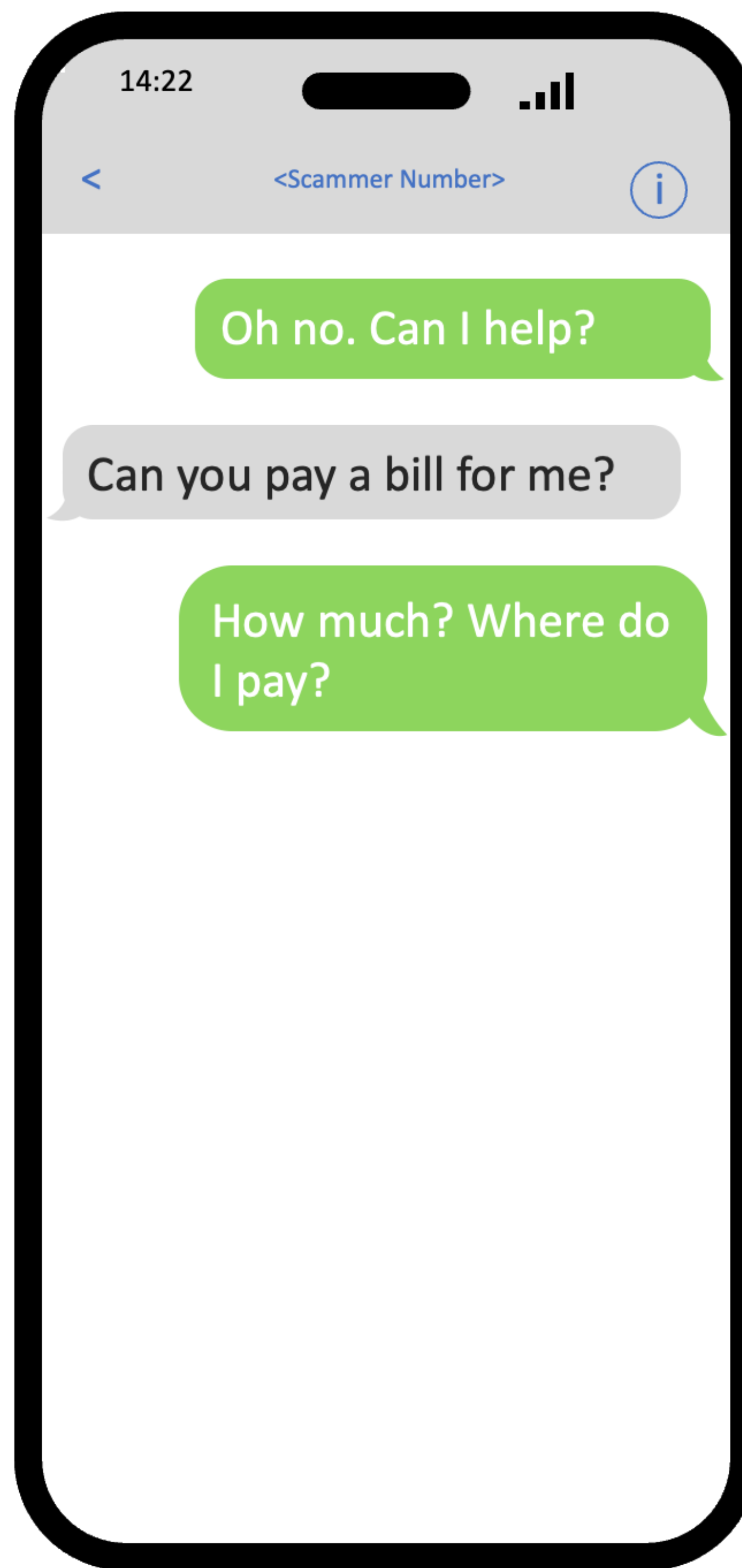
Scammer's first response

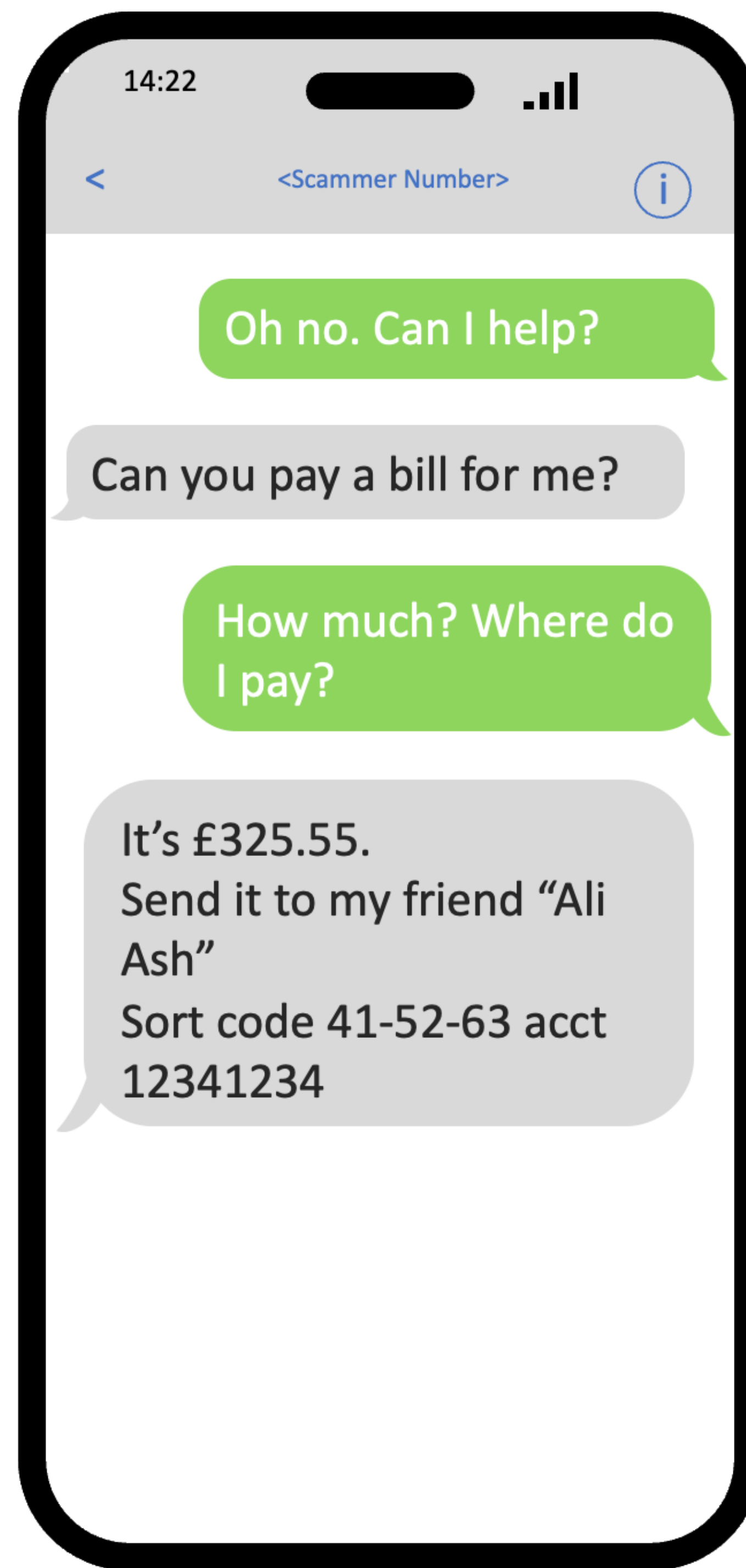




Scammer asks for help in
second response



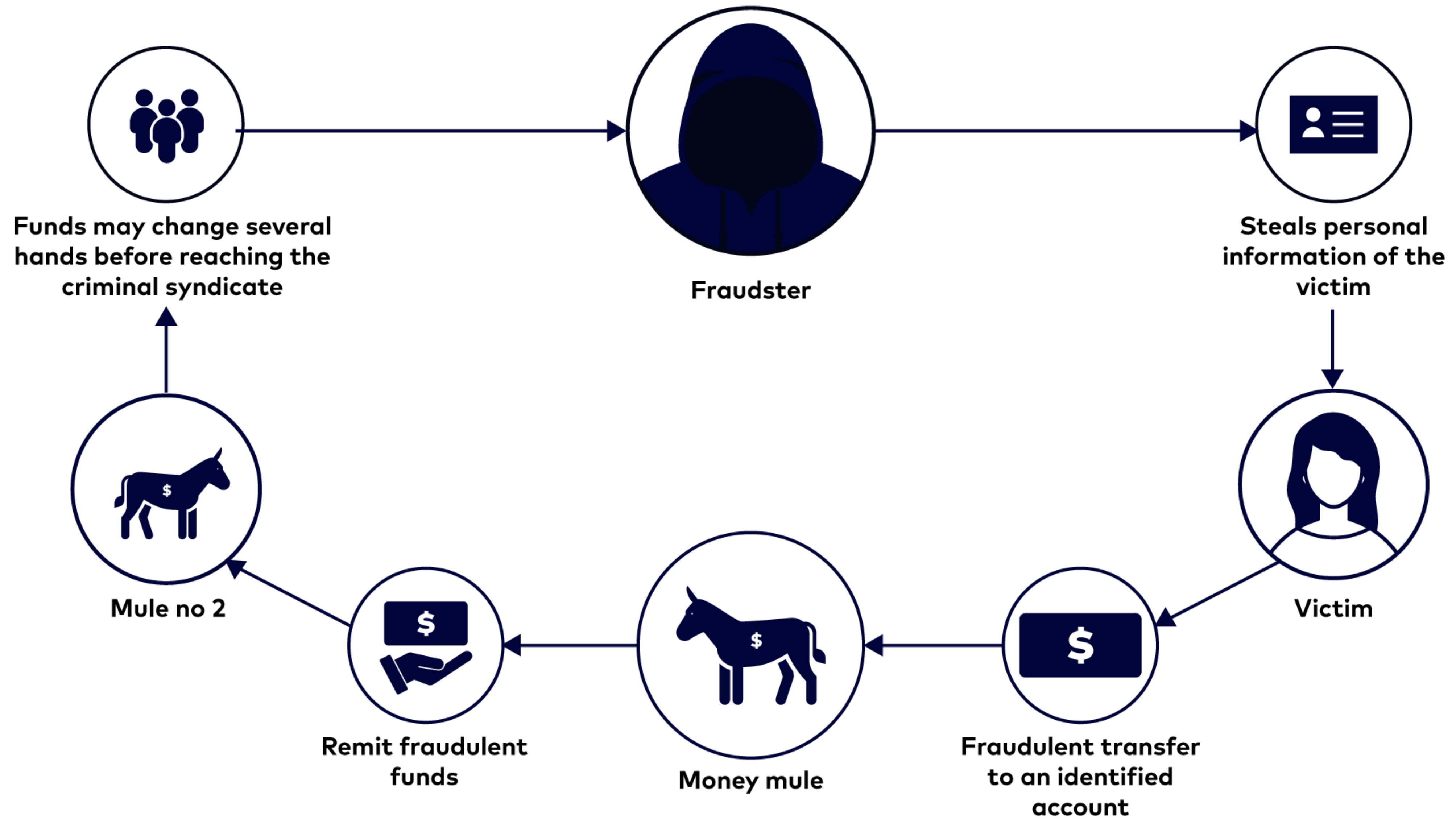




Scammer provides **suspect**
mule account

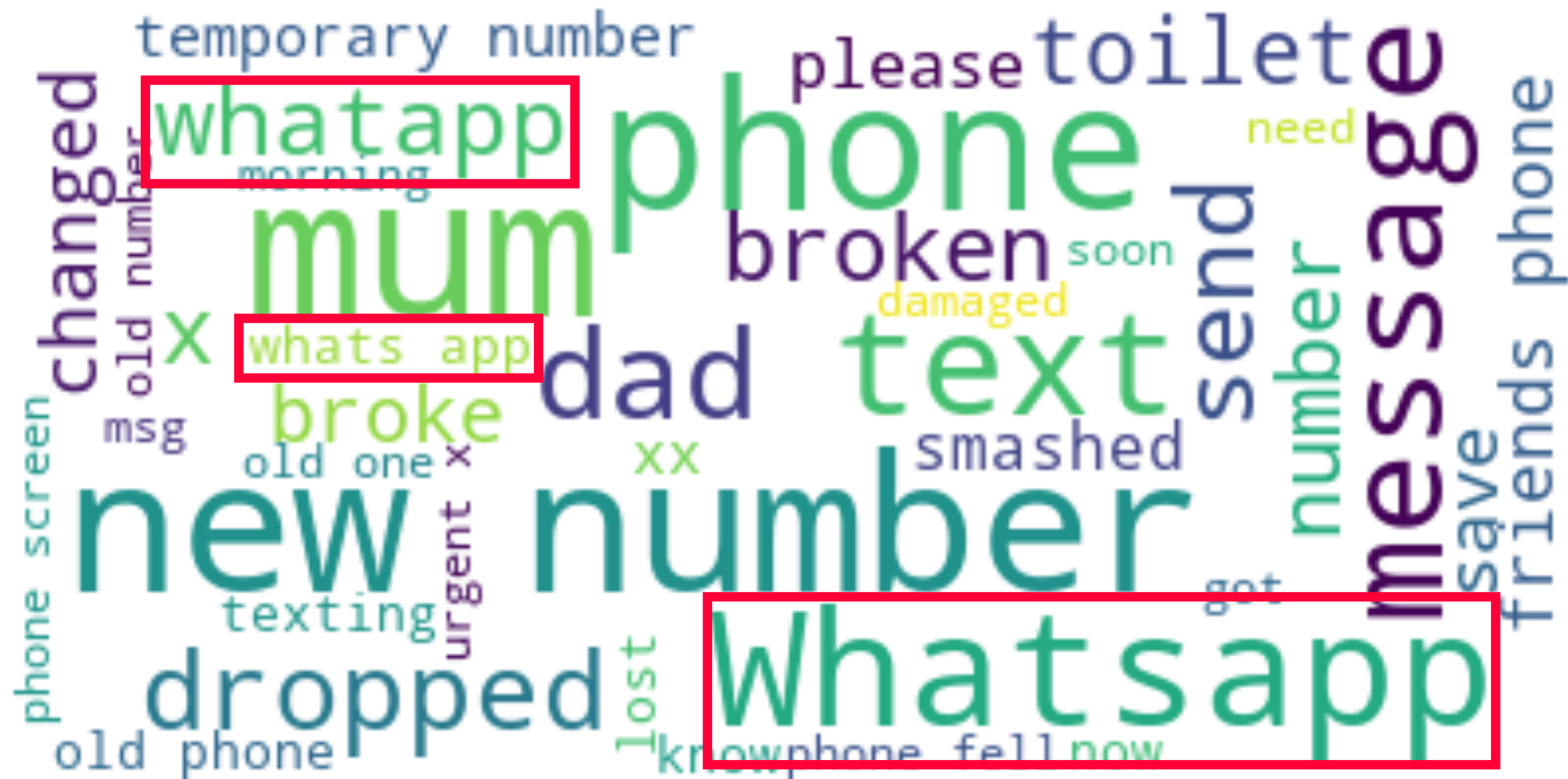
Money mule

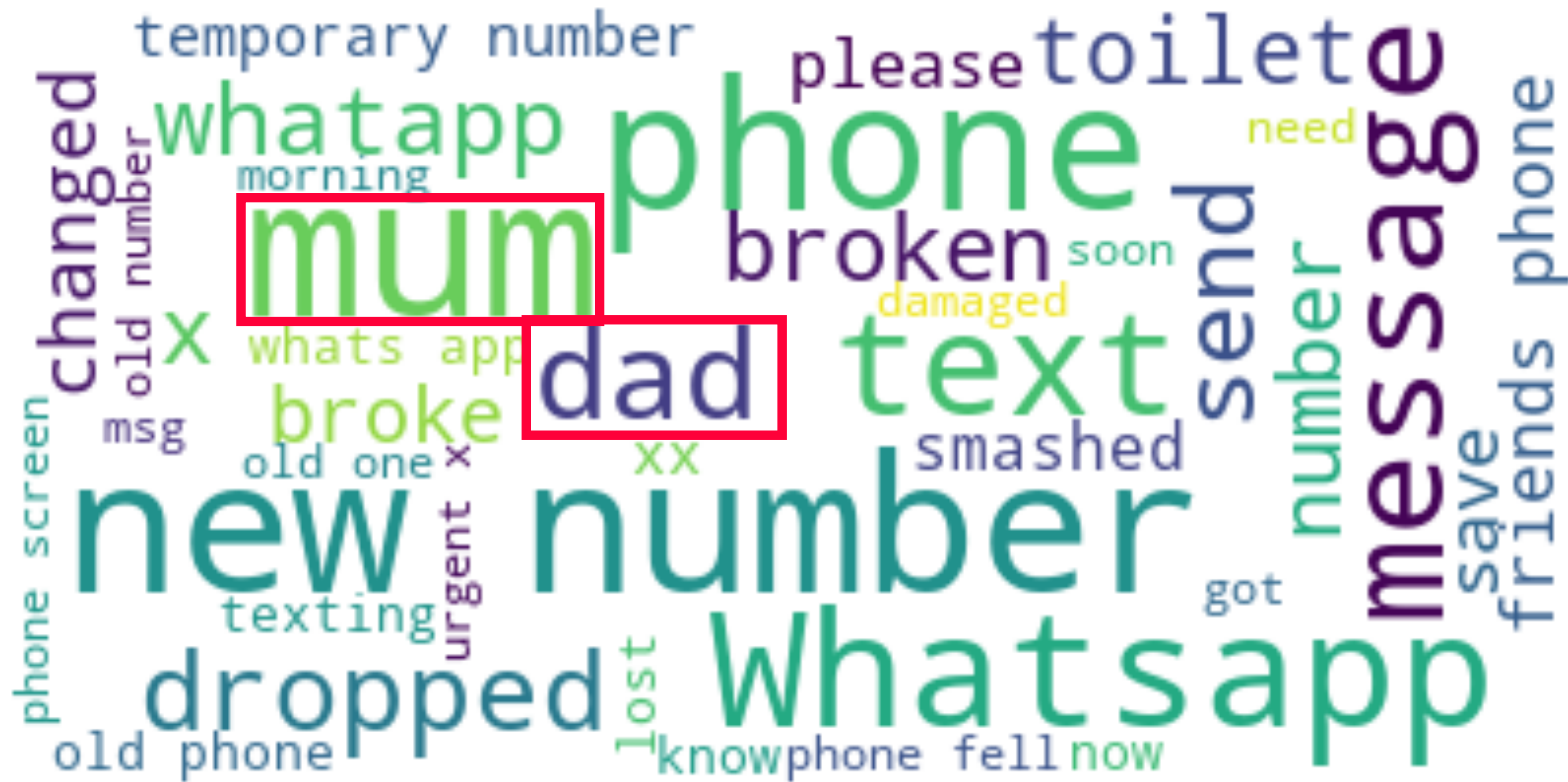
10|33



Methodology

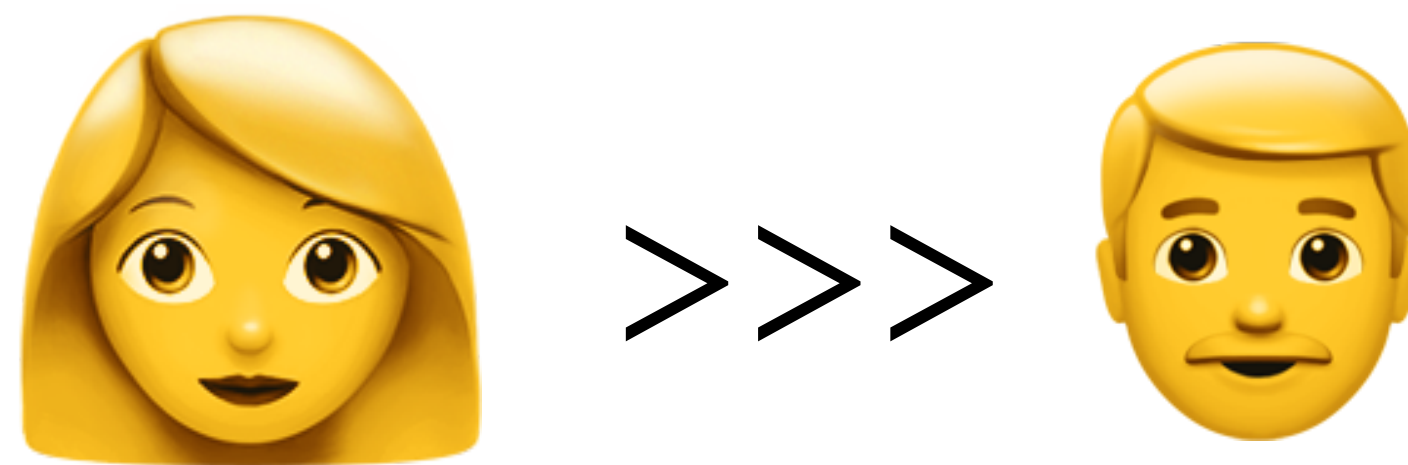
- Collaborate with **Stop Scams UK** and a **UK mobile network operator (MNO)**
- Textual analysis 💬 of **711 engaged scammer conversations** from **3,402 initial scam texts**
- Mobile network analysis 📱 of **486 originating sender IDs** and **1,184 scammer mobile numbers** using home location register (HLR) lookup 🔍
- Financial analysis 💰 of **582 unique suspect mule accounts** provided by scammers
 - Geographically locate 📍 **suspect mule accounts** associated with a **physical branch**
- Identify potential scammer networks 🕸 using **334 unique suspect mules** provided by **136 unique scammer numbers**





Results: Scammer strategies

Scammers prefer to target **female** users

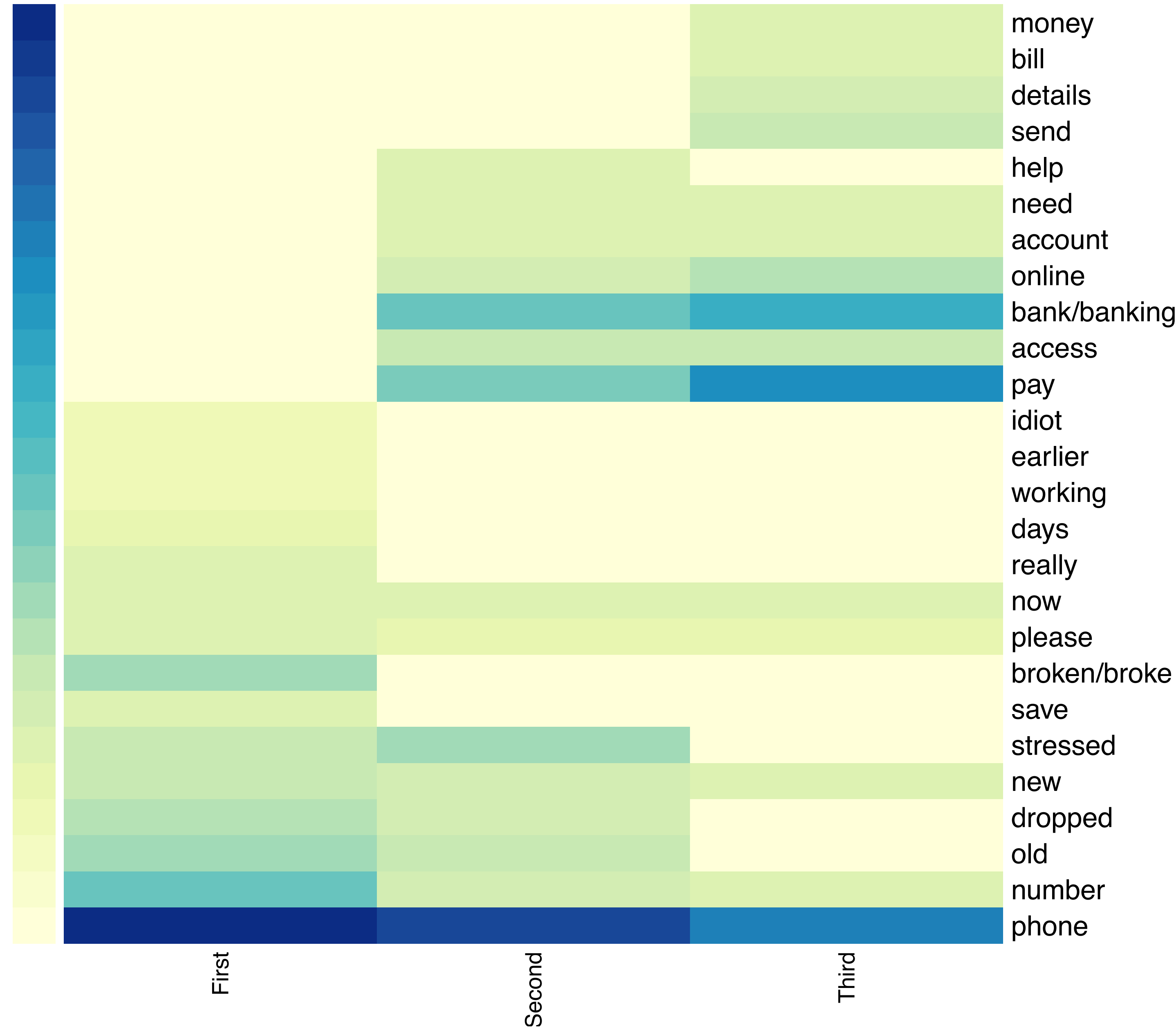


Results: Engagement

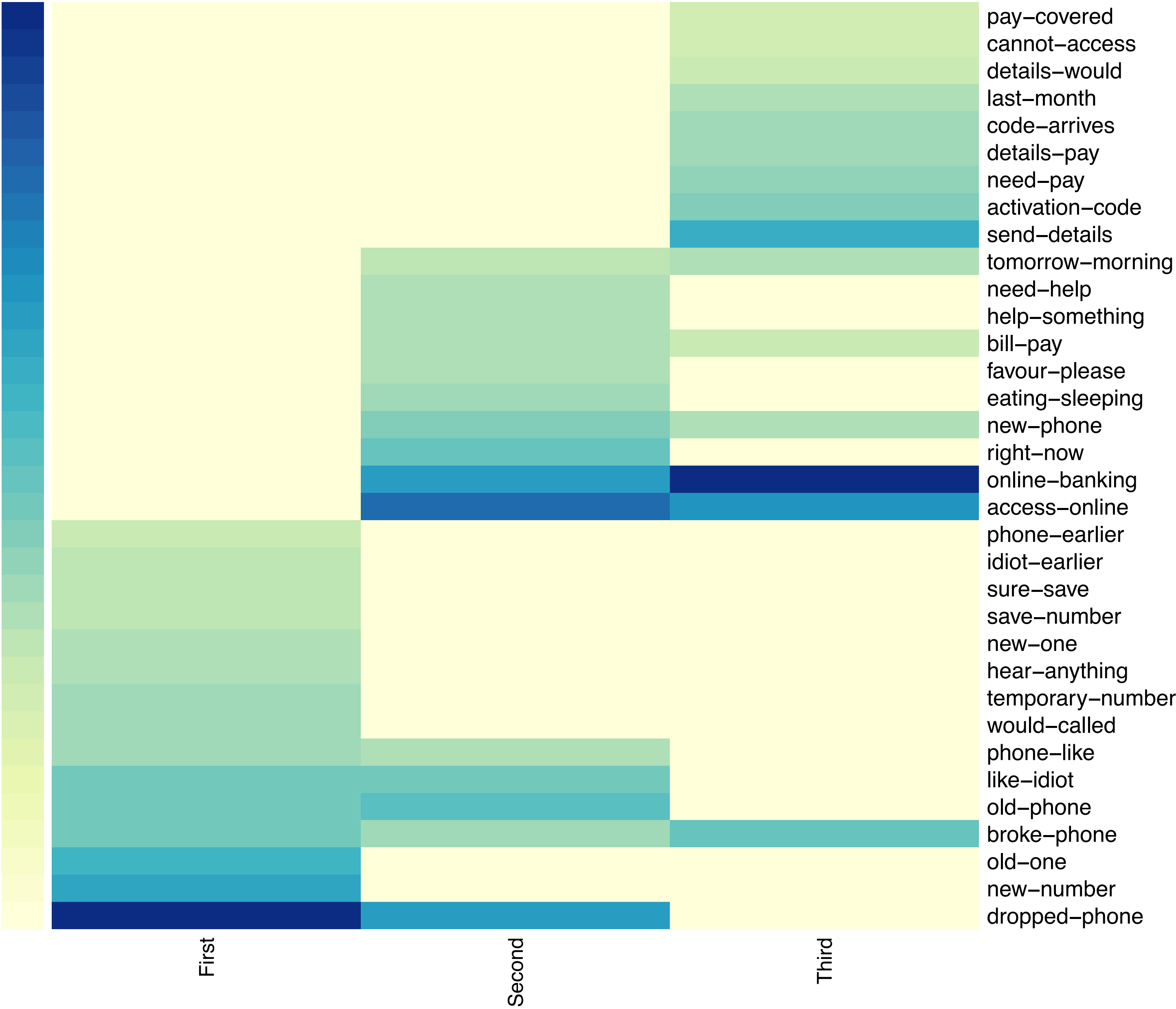
Platforms	Profiles	Mobile numbers used	Scammers contacted	Responses received	Response rate
SMS	Mum	10	881	116	13.17%
	Dad	10	514	63	12.26%
	No label	10	419	52	12.41%
Online messaging	-	100	2,313	167	7.22%

Number of unique scammers ($n=859$, $m=2,313$) contacted per profile and the initial responses received ($n=231$, $m=167$) over SMS and online messaging platforms

Scammer response: Unigram



Scammer response: Bigram



Scam lures

- Distraction - Confuse users by providing unrelated details.
- Kindness - Leverage willingness of people to help others.
- Time - Pressure users so they make an irrational decision.
- Social Compliance - Society trains people not to question authority.
- Herd - Participating as others share the same risks.
- Dishonesty - Inviting users willingly to a fraudulent scheme.
- Need and Greed - Leverage users' greed and offer attractive benefits

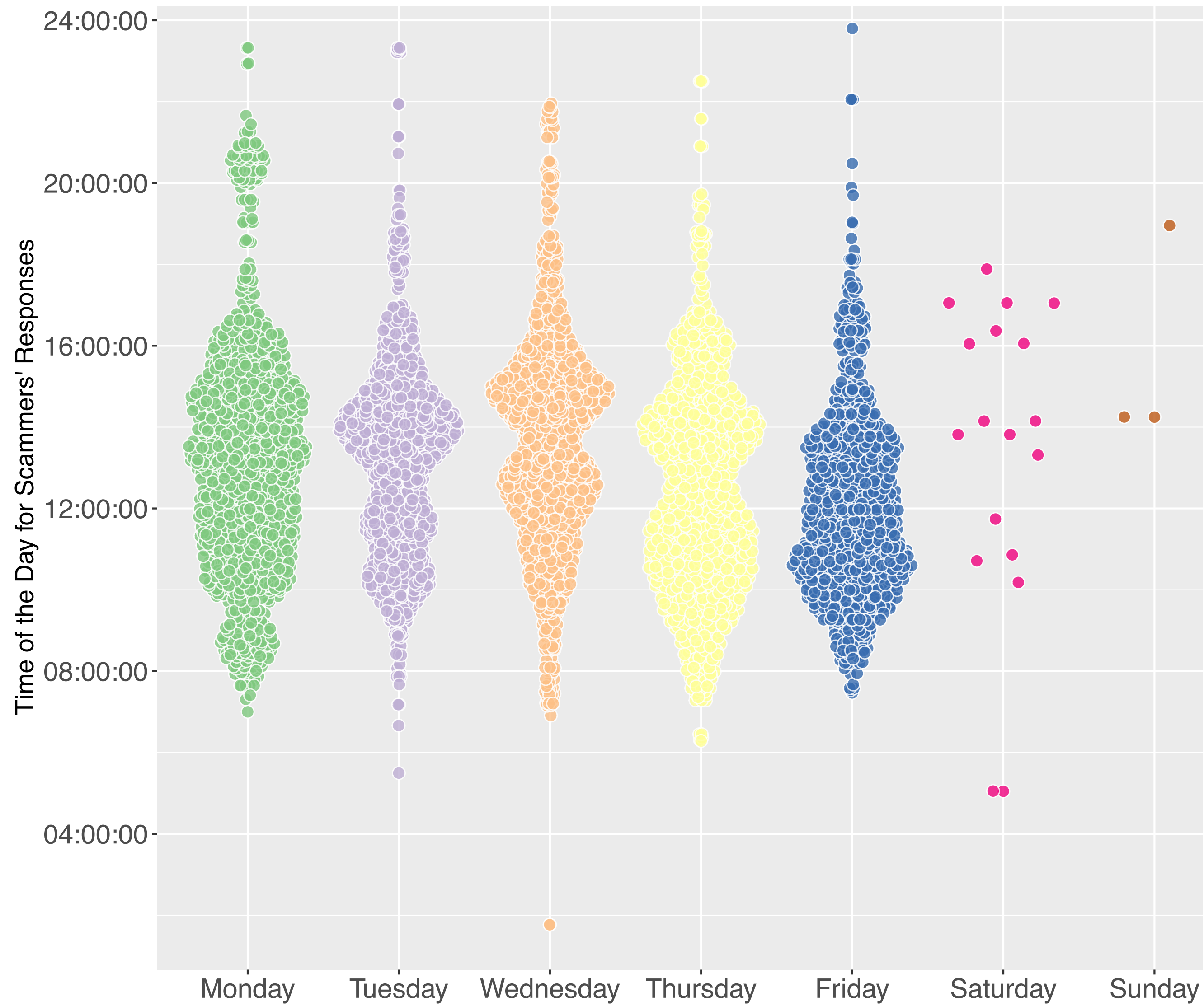


Results: Scammer strategies

Scammers use the **distraction, kindness, and time/urgency** lures to deceive users.



Scammer response: Timestamp



Time of the day per week when scammers respond to our messages ($n = 5,768$).

Home Location Register (HLR) Lookup

Check Mobile Numbers':

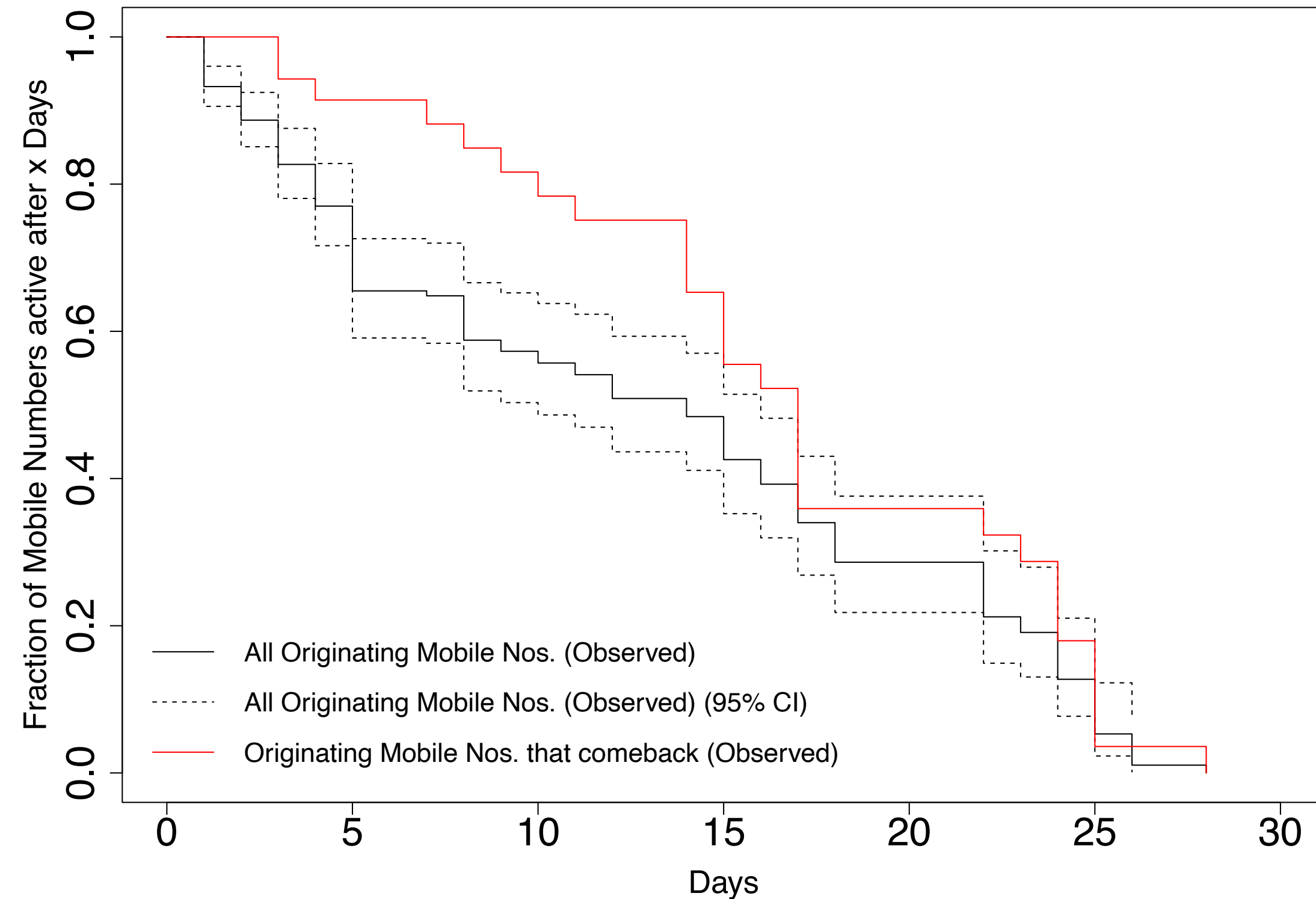
- Validity
- Availability and Status
- Network operator
- If it is ported
- If it is on roaming

TYPE OF NUMBER	Not Available
STATUS	NOT_AVAILABLE_NETWORK_ONLY
CURRENT NETWORK	Eir
CURRENT MCCMNC	27203
CURRENT COUNTRY	Ireland
CURRENT COUNTRY CODE	353
CURRENT COUNTRY ISO3	IRL

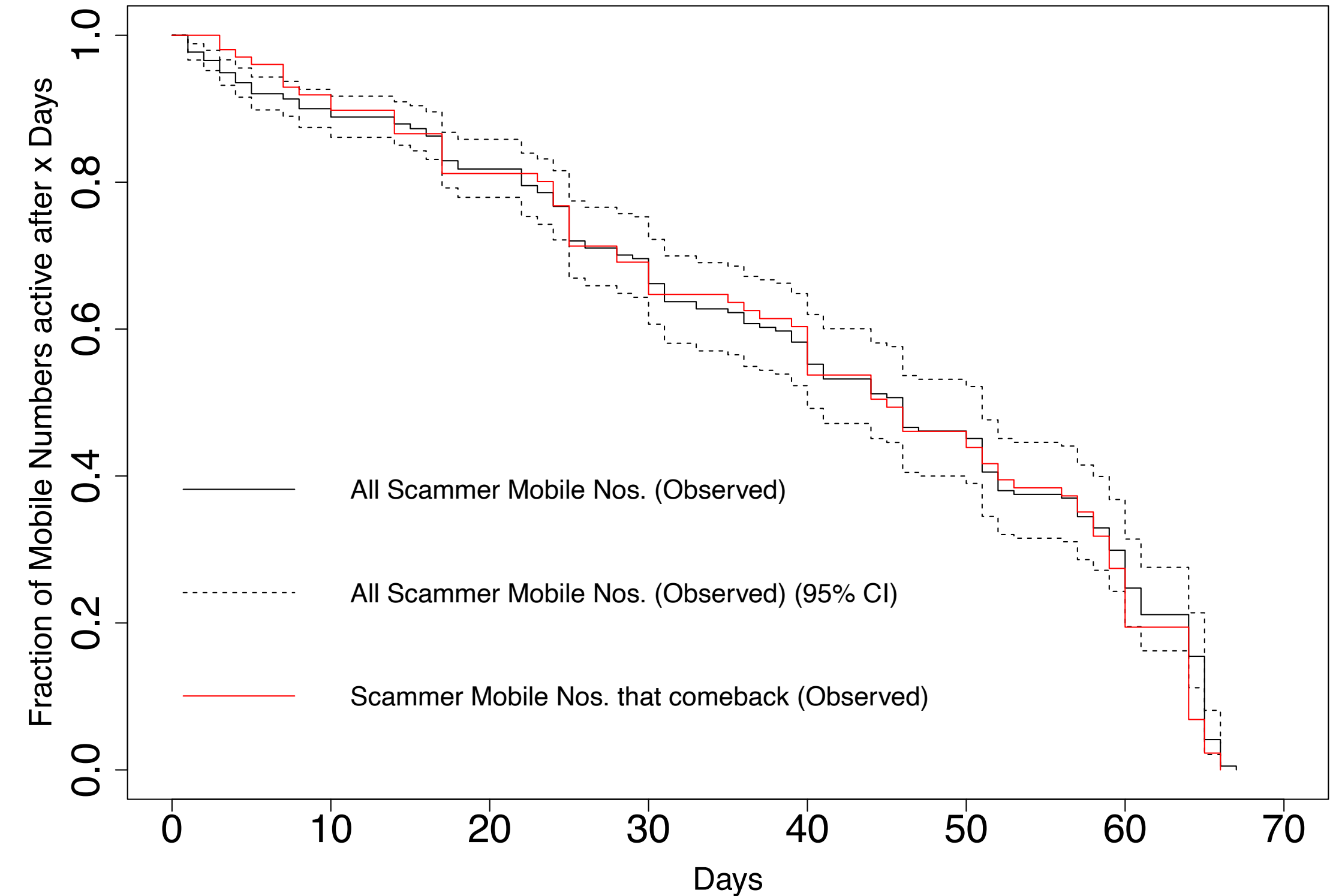
Mobile Network Operators (MNOs) Distribution

Mobile Network Operators	Type	SMS			Online Messaging	
		Originating <i>mobile phone numbers</i>	Scammer <i>mobile phone numbers</i>	Continuing <i>mobile phone numbers</i>	Originating <i>mobile phone numbers</i>	Scammer <i>mobile phone numbers</i>
MNO 1	Physical	326	601	321	-	96
MNO 2	Physical	114	272	76	-	30
MNO 3	Physical	23	150	8	-	22
MNO 4	Physical	8	81	2	-	14
MNO 5	Physical (MVNO)	1	18	0	-	3
MNO 6	Virtual	4	8	0	-	0
MNO 7	Virtual	0	8	0	-	2
MNO 8	Physical (MVNO)	10	4	1	-	0
MNO 9	Virtual	0	1	0	-	0
MNO 10	Virtual	0	1	0	-	0
MNO 11	Virtual	0	1	0	-	0
MNO 12	Virtual	0	1	0	-	0
MNO 13	Pager	0	1	0	-	0
MNO 14	Landline	0	1	0	-	0
Total		486	1,148	408	167	

Survival Analysis: Mobile network operators



Originating Sender ID mobile numbers ($n=486$), along with the mobile numbers that become active after being inactive ($n = 35$).



Scammer mobile numbers ($n=1,184$), along with the mobile numbers that become active after being inactive ($n = 101$).

Suspect Mules

- Over £577k  requested from 270 unique scammer mobile numbers 
- 582 unique suspect mules  identified that belong to over 30 financial institutions 

Platforms	Median	Mean	Min.	Max.	Requests
SMS	880	1,102	100	5,924	344
Online messaging	1,244	1,453	180	6,000	220
Total	£577,792				564

Distribution of all transaction requests (£).

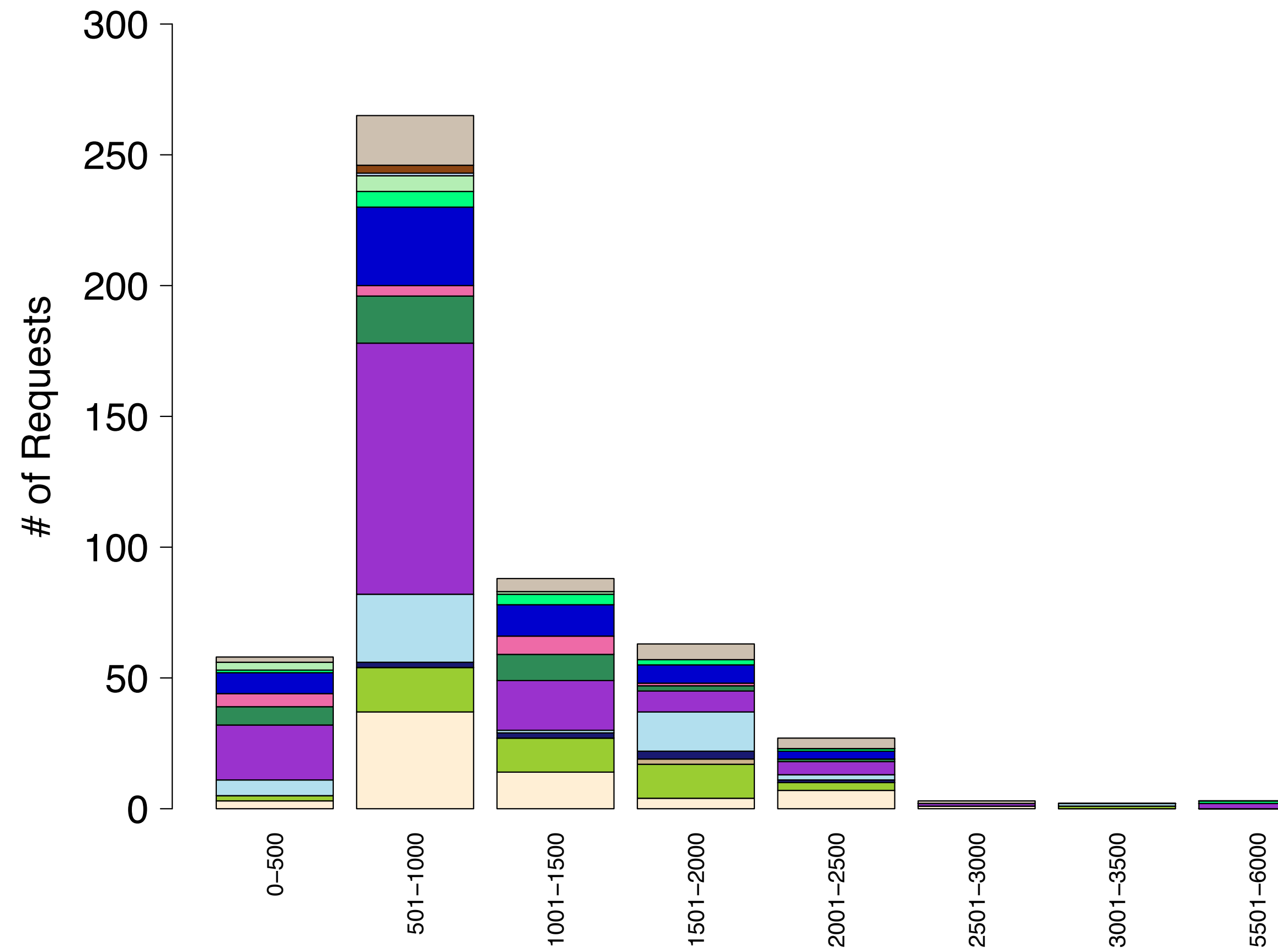
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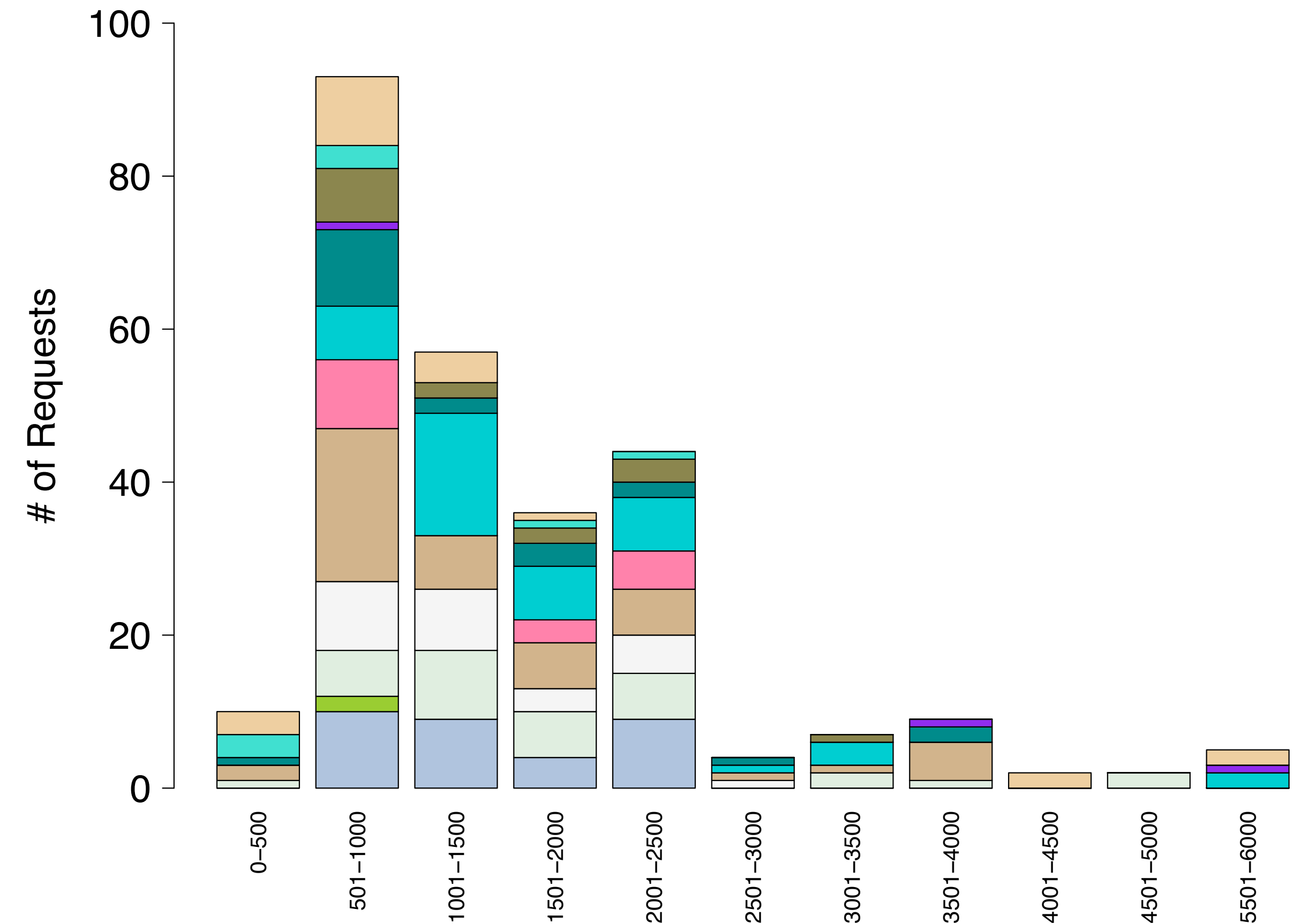
Category	Financial institutions	Mule accounts (unique)
Electronic Money Institution	14	374
High Street	12	196
EEA Authorised (International)	2	6
Authorised Payment Institution	5	5
Small Electronic Money	1	1

Distribution of unique mule accounts ($n=582$) aggregated with their respective category of financial institutions ($n = 34$) as per the UK's FCA handbook

Distribution of Requested amounts (£)

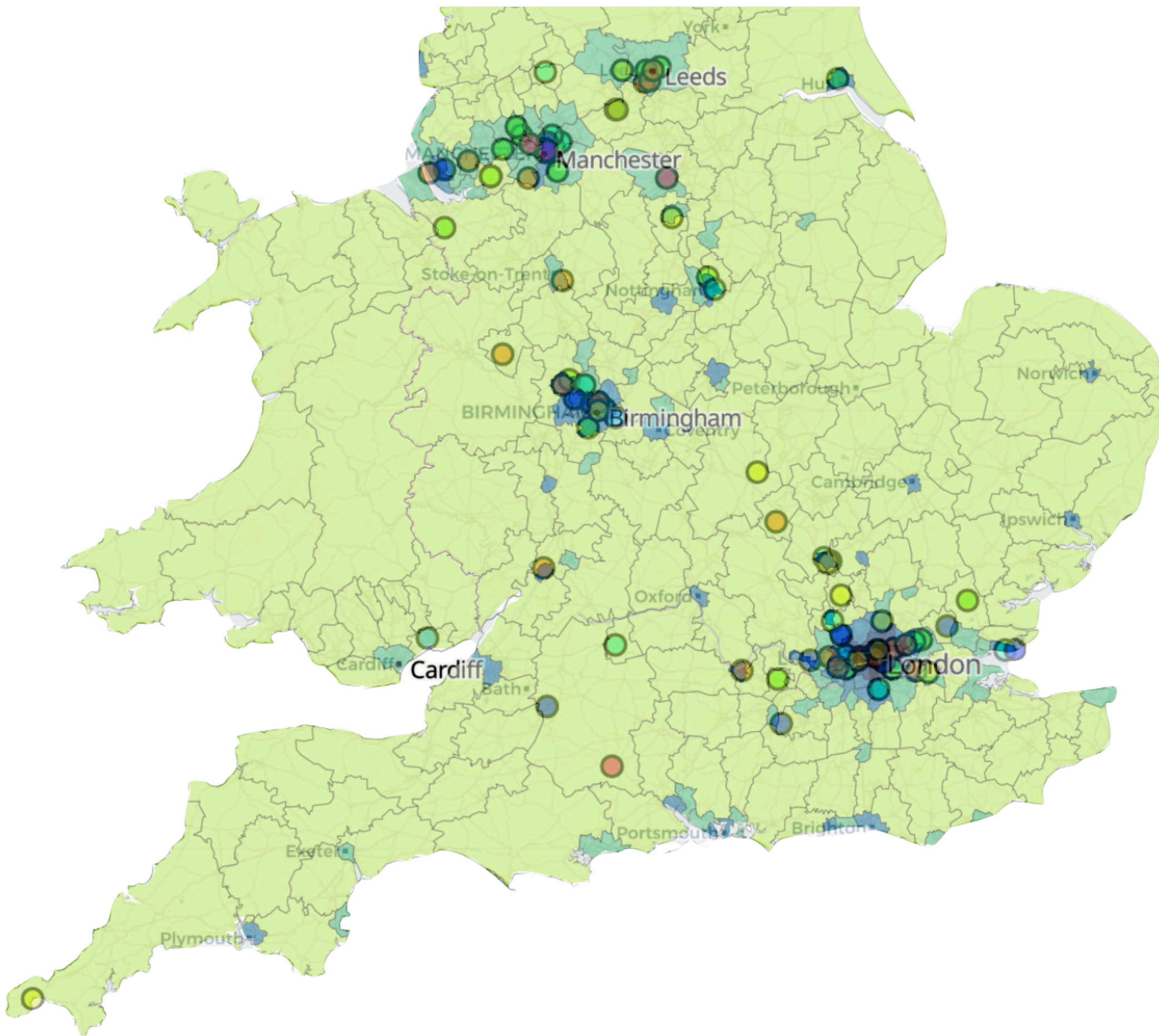


Electronic money institutions ($n=14$)
transaction requests ($n=509$)

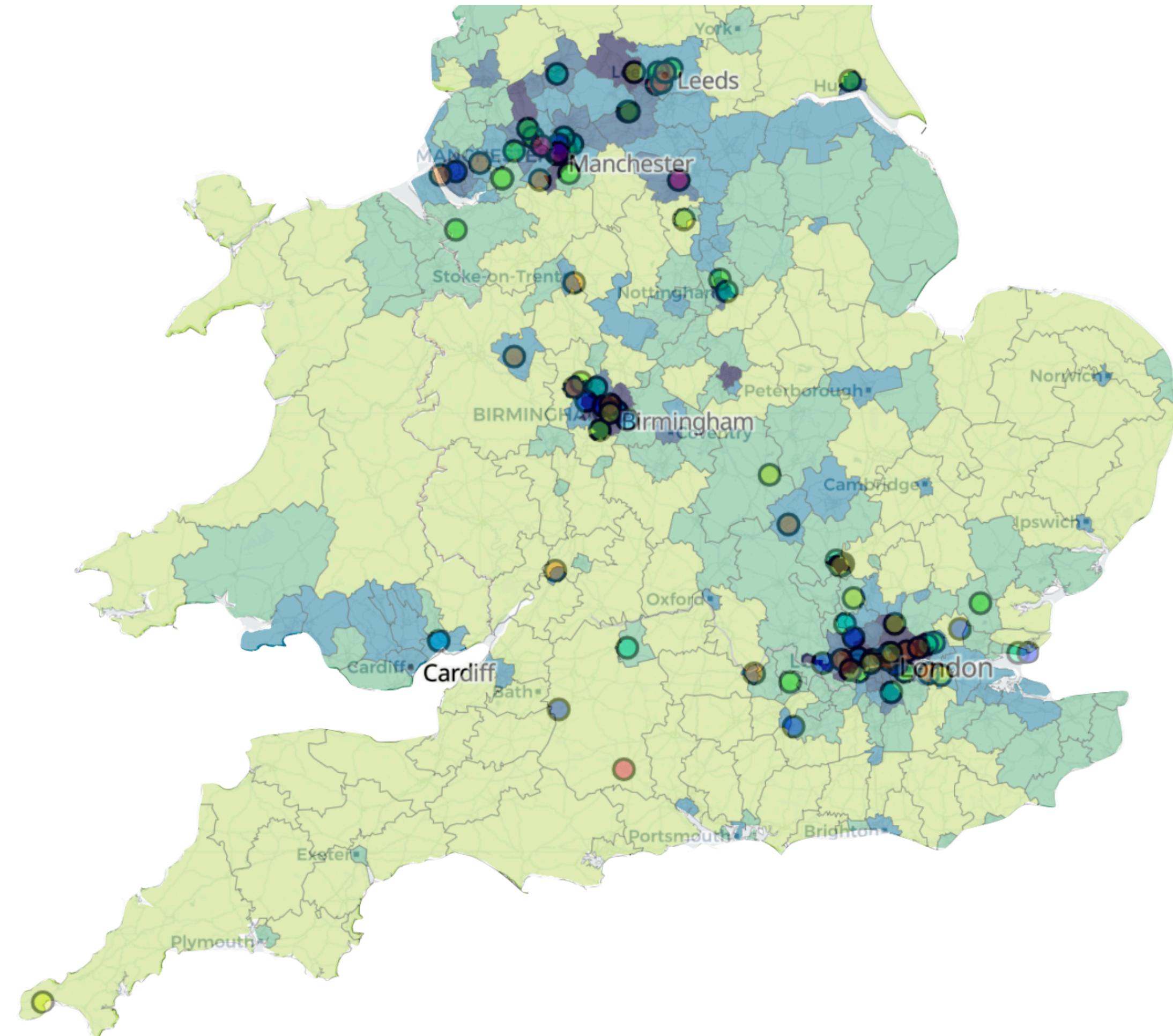


High street banks ($n=12$)
transaction requests ($n=269$)

Suspect mule geolocation



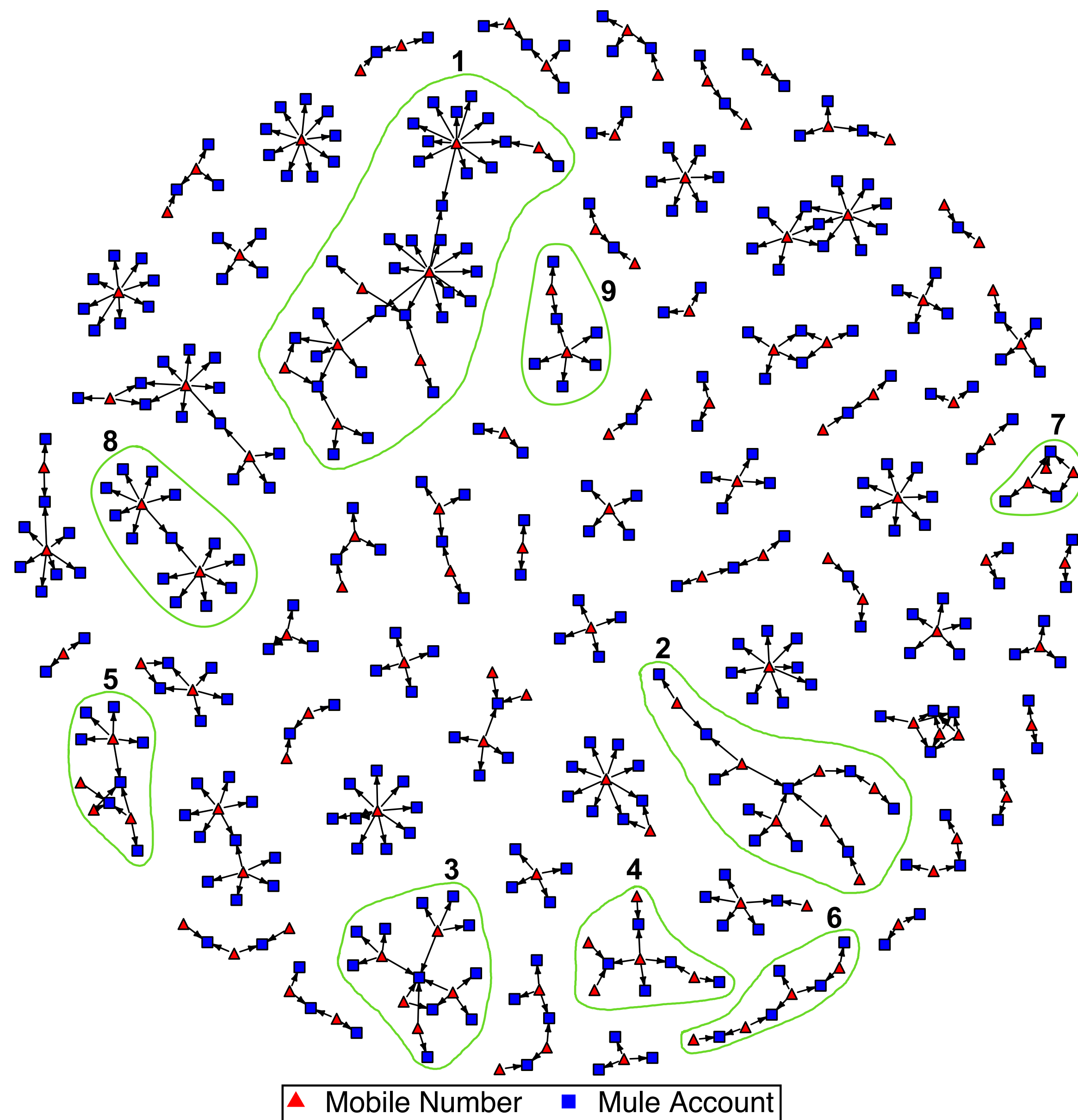
Population Density



Unemployment Rates

Suspect mule accounts ($n=96$) in High Street Banks ($n=12$)

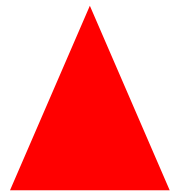
Scammer networks

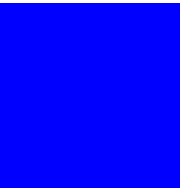


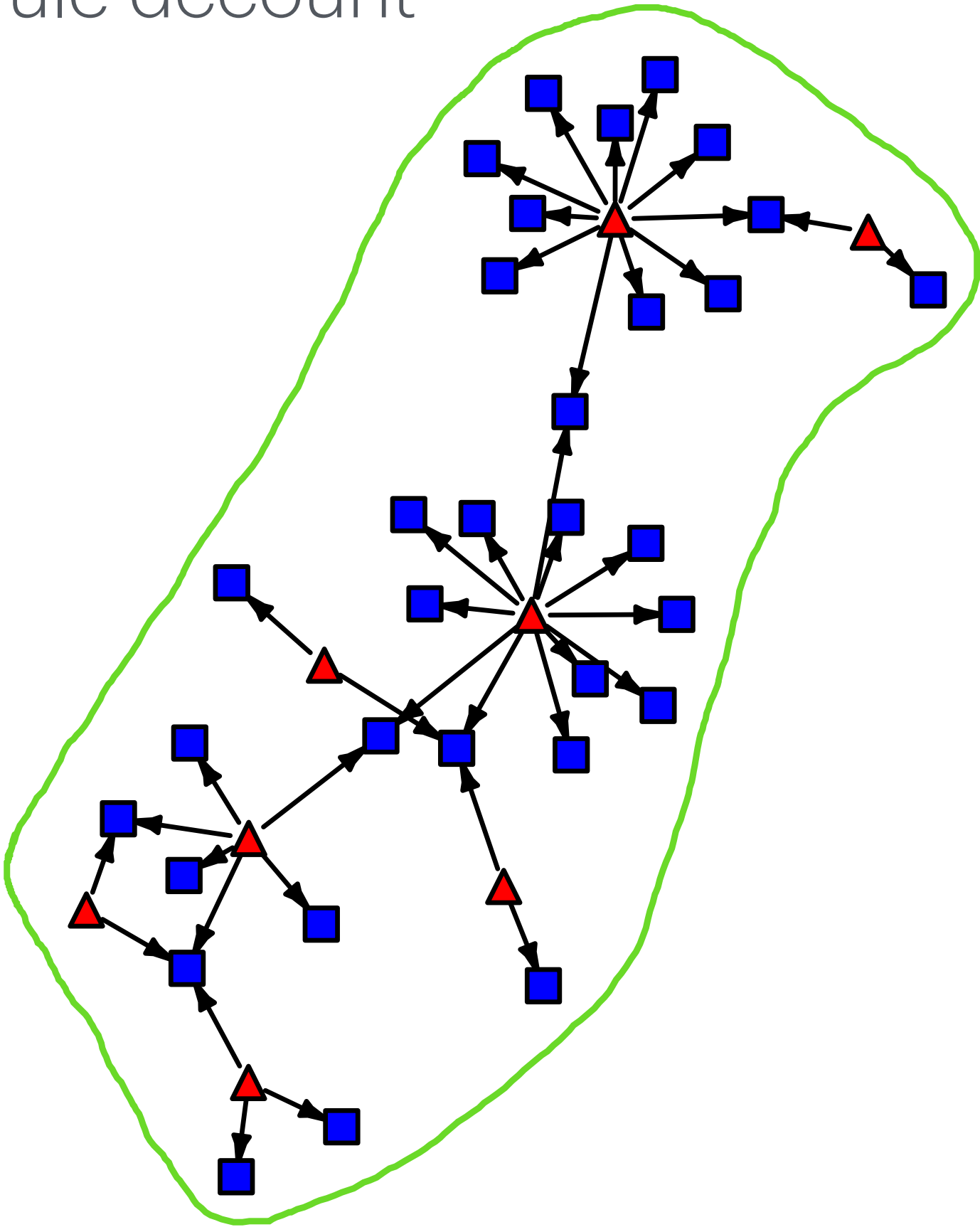
Group	Scammer mobile phone number	Mule accounts	Financial institutions	Amount requested (£)
1	8	31	8	41,125
2	7	10	7	14,487
3	5	11	4	9,452
4	5	5	5	11,400
5	4	7	6	14,231
6	4	5	4	11,728
7	3	3	3	18,000
8	2	13	7	10,700
9	2	6	6	10,965

Breakdown of amounts requested (round-off to nearest decimal) by the nine communities identified.

Scammer networks: Group 1

 Scammer mobile number

 Mule account



Scammer Phone Numbers	Suspect Mule Accounts	Financial Institutions	Amount Requested
8	31	8	£41,125

Results submitted as consultation responses



Consultation response form

Please complete this form in full and return to mobilemessagingcamsresponses@ofcom.org.uk.

Consultation title	Call for input: Reducing mobile messaging scams
Full name	Mr. Sharad Agarwal and Dr. Marie Vasek
Contact phone number	[REDACTED]
Representing (delete as appropriate)	Self
Organisation name	Information Security Research Group, Computer Science Department, University College London (UCL)
Email address	[REDACTED] and [REDACTED]

Spanish Government Launches Consultation on Tackling Telecom Scams

By Eric Priezkalns 19 Feb 2024 Risk, Fraud & Security

Iniciativas y Mecanismos Regulatorios Técnicos y Operativos para Combatir Estafas

Respuesta a la Consulta Pública

Presentada por el Sr. Sharad Agarwal y la Dra. Marie Vasek del Grupo de Investigación en Seguridad de la Información en *University College London (UCL)*, Reino Unido, y el Dr. Guillermo Suárez-Tangil de *IMDEA Networks*

May 4, 2024

<https://sharad1126.github.io>

Impact: New legislation in Spain to tackle SMS scams



Agencia Estatal Boletín Oficial del Estado

Consolidated legislation

[Aid](#)

Order TDF/149/2025, of February 12, establishing measures to combat identity theft scams through fraudulent phone calls and text messages and to ensure the identification of the numbers used to provide customer service and make unsolicited commercial calls.

Published in: «BOE» No. 40, of 15/02/2025.
Entry into force: 07/03/2025
Department: Ministry for Digital Transformation and Public Service
Reference: [BOE-A-2025-2870](#)
Permalink ELI: <https://www.boe.es/eli/es/o/2025/02/12/tdf149/con>

Select wording: Initial text published on 02/15/2025

[PDF](#)

[ePUB](#)

Takeaways

- Users in the UK lose at least **£2.3m (\$2.9m)** per year to 'Hi mum and dad' scams. This is a growing concern around the world, including the US.
- Scammers prefer mobile network operators that support GSM SIM boxes/banks/farms.
- Scammers continuously try to identify the amounts that could evade detection from financial institutions.
- Network analysis on suspect mules and mobile numbers indicate possible clusters (threat actor groups).

Read our paper:



sharad.agarwal@ucl.ac.uk

<https://sharad1126.github.io>

Thank you!

Results: Scammer strategies

Yes **mum**, I **dropped** it like an **idiot**. This is my **new number**....

First



Distraction &
Kindness

You should be more careful, love. Are you ...

Im **stressed**. I **need** to make a **payment** and my **bank** wouldn't let me

Second



Kindness

Don't stress, are you taking your phone for a repair?

I cannot **access** my **bank** and I need to **urgently pay** £1,765. Could you **please pay** it **now** and I promise to send it back as soon as my **bank** allows?

Third



Kindness &
Time/Urgency